

Congruence Classes of Triangles in a Cubical Grid

Exact encoding and sharp order of growth for OEIS A400663

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Abstract

For $n \geq 0$, let $G_n = \{0, 1, \dots, n\}^3$, and let $a(n)$ be the number of Euclidean congruence classes of nondegenerate triangles with vertices in G_n . This is the three-dimensional analogue of the square-grid enumeration A028419. We give an exact finite encoding using $3n(n+1)/2+1$ one-coordinate contribution types, prove strict monotonicity, and determine the sharp order of growth:

$$a(n) \asymp \frac{n^6}{\sqrt{\log n}}.$$

The logarithmic factor comes from local quadratic conditions on the Gram determinant. Along the way we show that a primitively representable integral Gram matrix of determinant D has exactly $24\rho(D)$ ordered primitive representations in $\mathbb{Z}^3$, where $\rho(D)$ is the number of square roots of 1 modulo D . Consequently,

$$\log a(n) = 6 \log n - \frac{1}{2} \log \log n + O(1).$$

No computation is used in the proofs.

1 Introduction and motivation

Finite lattice grids provide a simple setting in which a geometric counting problem quickly acquires arithmetic structure. With the same parameter convention used here, A028419 counts congruence classes of triangles with vertices in the planar grid $\{0, 1, \dots, n\}^2$; related OEIS sequences count similarity classes (A028492) and split the congruence classes into acute, right, and obtuse triangles (A190021, A189979, A190022); A241231 treats a centered hexagonal grid [7]. The natural three-dimensional analogue is to replace the square grid by the cubical grid

$$G_n = \{0, 1, \dots, n\}^3$$

and to identify triangles only by ordinary Euclidean congruence. Thus two triangles are identified exactly when their three side lengths agree; no lattice-preserving motion is required.

The resulting sequence begins

$$0, 3, 40, 214, 830, 2355, 6141, 13566, 27369, \dots$$

and is OEIS A400663. An independent online enumeration reported the first eight terms and their acute/right/obtuse decomposition on 13 April 2026 [6]; the present work gives the exact encoding and the asymptotic results below. The first nontrivial question is not how many placements occur, but how many different shapes occur. A triangle is determined by its three side lengths, so the geometry can be compressed to a set of integer triples of squared lengths.

The key observation is that each coordinate direction contributes independently to those three squared lengths. This reduces a three-dimensional congruence problem to a finite arithmetic sumset problem.

The first values are

n	0	1	2	3	4	5	6	7	8
$a(n)$	0	3	40	214	830	2355	6141	13566	27369

A companion illustration in the Zenodo record displays all 40 congruence classes for $n = 2$. The displayed values are computed from the exact encoding below and are not used in any proof.

The paper has two layers. Sections 2 and 3 are elementary: they give the exact encoding, an $O(n^6)$ upper bound, and strict monotonicity. Section 4 identifies the missing logarithmic factor and proves the sharp order $a(n) \asymp n^6/\sqrt{\log n}$. The geometric idea remains simple: primitive edge pairs are grouped by their Gram matrices. The arithmetic input is that primitive representations can be counted exactly and that the admissible Gram matrices satisfy a half-dimensional family of local conditions. The technical details are kept in the appendices.

2 The one-coordinate encoding

A triangle is determined up to Euclidean congruence by its three side lengths, hence also by the sorted triple of its squared side lengths.

Suppose one vertex is translated to the origin and the other two edge vectors are $u, v \in \mathbb{Z}^3$. In one coordinate we see three integers $0, r, s$. Their contribution to the three squared side lengths is

$$(r^2, s^2, (r-s)^2).$$

The three points can be translated into $\{0, \dots, n\}$ in that coordinate exactly when their span is at most n . This motivates

$$C_n = \{(r^2, s^2, (r-s)^2) : r, s \in \mathbb{Z}, \max(0, r, s) - \min(0, r, s) \leq n\}.$$

Thus C_n records precisely the possible *one-coordinate contributions* to the three squared side lengths. Equivalently, it is the image of

$$(a, b, c) \mapsto ((b-a)^2, (c-a)^2, (b-c)^2)$$

on $\{0, 1, \dots, n\}^3$.

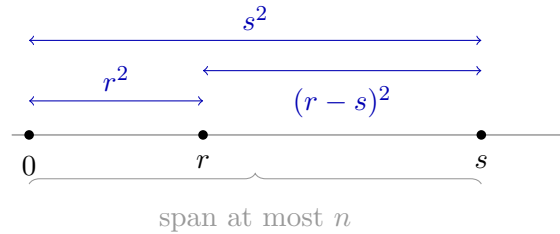


Figure 1: One coordinate contributes the three squared differences r^2 , s^2 , and $(r-s)^2$. The span condition is exactly what permits translation into $\{0, \dots, n\}$.

Lemma 1. *One has*

$$|C_n| = \frac{3n(n+1)}{2} + 1.$$

Proof. The condition on (r, s) is equivalent to

$$|r| \leq n, \quad |s| \leq n, \quad |r - s| \leq n.$$

The admissible pairs are the lattice points of the hexagon cut out by these three inequalities. There are

$$(2n + 1) + 2 \sum_{r=1}^n (2n - r + 1) = 3n^2 + 3n + 1$$

such ordered pairs, the centered hexagonal number of radius n . Moreover, the triple $(r^2, s^2, (r - s)^2)$ determines rs through

$$2rs = r^2 + s^2 - (r - s)^2,$$

so two admissible pairs give the same element of C_n exactly when they are (r, s) and $(-r, -s)$. The involution has the single fixed point $(0, 0)$, and therefore

$$|C_n| = 1 + \frac{3n^2 + 3n}{2}.$$

□

For $x, y, z \geq 0$ put

$$H(x, y, z) = 2(xy + xz + yz) - x^2 - y^2 - z^2.$$

The polynomial H is symmetric, so the condition $H > 0$ is unaffected by sorting the three squared side lengths.

Theorem 1. *The sequence is given exactly by*

$$a(n) = \#\left\{\text{sort}(c_1 + c_2 + c_3) : c_1, c_2, c_3 \in C_n, H(c_1 + c_2 + c_3) > 0\right\},$$

where addition is componentwise.

Proof. Let the vertices of a triangle be $p_0, p_1, p_2 \in G_n$, and put

$$u = p_1 - p_0, \quad v = p_2 - p_0.$$

For each coordinate i , the three integers $0, u_i, v_i$ have span at most n . The three squared side lengths are

$$\|u\|^2, \quad \|v\|^2, \quad \|u - v\|^2,$$

and the contribution of coordinate i is therefore

$$(u_i^2, v_i^2, (u_i - v_i)^2) \in C_n.$$

This proves that every occurring triple has the stated form.

Conversely, suppose three elements of C_n are represented by coordinate pairs (r_i, s_i) , $i = 1, 2, 3$. For each coordinate choose

$$t_i = -\min(0, r_i, s_i).$$

The span condition implies that $t_i, t_i + r_i, t_i + s_i$ all lie in $\{0, \dots, n\}$. Hence the three points with coordinate triples $t, t + r, t + s$ belong to G_n and have the required squared side lengths.

Finally, if the squared side lengths are x, y, z , then for two edge vectors u, v with $\|u\|^2 = x$, $\|v\|^2 = y$, $\|u - v\|^2 = z$ one has

$$2u \cdot v = x + y - z.$$

Hence the squared Heron identity gives

$$16A^2 = 4(\|u\|^2\|v\|^2 - (u \cdot v)^2) = 4xy - (x + y - z)^2 = H(x, y, z),$$

where A is the area. Thus $H > 0$ is exactly the nondegeneracy condition; degenerate or zero-length cases automatically have $H \leq 0$. □

Example: the three classes for $n = 1$. For $n = 1$,

$$C_1 = \{(0, 0, 0), (1, 0, 1), (0, 1, 1), (1, 1, 0)\}.$$

Adding three elements of C_1 componentwise, sorting, and retaining only triples with $H > 0$ gives exactly

$$(1, 1, 2), \quad (1, 2, 3), \quad (2, 2, 2),$$

so $a(1) = 3$. Figure 2 shows one representative of each class.

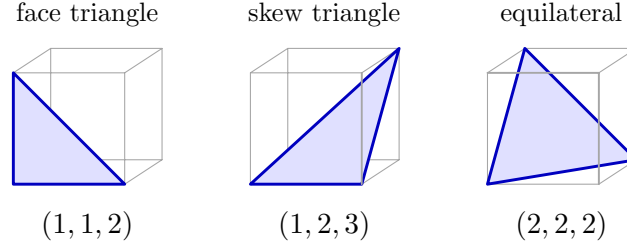


Figure 2: Representatives of the three congruence classes in G_1 . Labels give the sorted squared side lengths.

As an immediate consequence of the encoding, every occurring triple satisfies

$$x + y + z \equiv 0 \pmod{2},$$

since each one-coordinate contribution has sum $2(r^2 + s^2 - rs)$.

3 Elementary bounds and monotonicity

By Lemma 1, the number of multisets of three elements of C_n is

$$\binom{|C_n| + 2}{3} = \binom{3n(n+1)/2 + 3}{3}.$$

Thus

$$a(n) \leq \binom{3n(n+1)/2 + 3}{3} = O(n^6). \quad (1)$$

The same count shows that Theorem 1 gives a direct finite enumeration by inspecting $O(n^6)$ multisets of three contribution types.

The planar grid $\{0, 1, \dots, n\}^2$ embeds as a coordinate face of G_n , so the two-dimensional analogue A028419 gives

$$\text{A028419}(n) \leq a(n).$$

Since $G_{n-1} \subset G_n$, one has $a(n-1) \leq a(n)$. The inequality is in fact strict. For $n \geq 1$ the triangle

$$(0, 0, 0), \quad (n, 0, 0), \quad (n, n, n)$$

has squared side lengths $n^2, 2n^2, 3n^2$, and therefore diameter $\sqrt{3}n$. Every pair of points of G_{n-1} is at distance at most $\sqrt{3}(n-1)$, so this congruence class is new at level n . Hence

$$a(n) > a(n-1) \quad (n \geq 1).$$

4 Sharp order of growth

The elementary upper bound in Section 3 has order n^6 , but the true count is smaller by a square root of a logarithm. The reason is arithmetic. A triangle with edge vectors $u, v \in \mathbb{Z}^3$ has Gram matrix

$$G(u, v) = \begin{pmatrix} \|u\|^2 & u \cdot v \\ u \cdot v & \|v\|^2 \end{pmatrix}, \quad D = \det G(u, v) = \|u \times v\|^2.$$

At every odd prime dividing D , primitive representability imposes one quadratic-residue condition. These local conditions accumulate with sieve dimension $1/2$.

Call an ordered pair (u, v) *primitive* if the rank-2 lattice $\mathbb{Z}u + \mathbb{Z}v$ is primitive in $\mathbb{Z}^3$, equivalently if the three coordinates of $u \times v$ are coprime. For a positive integer D put

$$\rho(D) = \#\{t \bmod D : t^2 \equiv 1 \pmod{D}\}, \quad \rho(1) = 1.$$

Proposition 1 (Exact primitive multiplicity). *Let*

$$G = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$$

be a positive definite integral matrix of determinant $D = ab - c^2$. If G has a primitive representation

$$G = G(u, v), \quad u, v \in \mathbb{Z}^3,$$

then the number of ordered primitive representations is exactly

$$24\rho(D).$$

In particular, since $4 \nmid D$ for a primitive representation,

$$24\rho(D) = \begin{cases} 24 \cdot 2^{\omega(D)}, & D \text{ odd}, \\ 12 \cdot 2^{\omega(D)}, & D \equiv 2 \pmod{4}. \end{cases}$$

The proof is given in Appendix A. That appendix also gives a local criterion for primitive representability. The point needed here is that Proposition 1 turns the variable multiplicity of one Gram matrix into the simple weight $1/\rho(D)$.

For $m \geq 2$ define

$$W(m) = \sum_{\substack{u, v \in [-m, m]^3 \\ \mathbb{Z}u + \mathbb{Z}v \text{ primitive} \\ u \times v \neq 0}} \frac{1}{\rho(\|u \times v\|^2)}.$$

Proposition 2 (Weighted primitive pairs). *As $m \rightarrow \infty$,*

$$W(m) \gg \frac{m^6}{\sqrt{\log m}}.$$

The proof is in Appendix B. Its local input is particularly simple. For an odd prime p , among rank-2 pairs modulo p the proportion for which

$$p \mid \|u \times v\|^2$$

is

$$h(p) = \frac{p+1}{p^2+p+1} = \frac{1}{p} + O(p^{-3}). \quad (2)$$

The factor $1/\rho(D)$ contributes $1/2$ when such an odd prime divides D . Thus the relevant Euler factors are

$$1 - \frac{1}{2p} + O(p^{-2}),$$

which is the source of the exponent $1/2$ in Proposition 2.

For the matching upper bound we use the same local obstruction from the opposite direction.

Proposition 3 (Half-dimensional upper bound). *Let $N(X)$ be the number of positive definite integral matrices*

$$\begin{pmatrix} a & c \\ c & b \end{pmatrix}, \quad 1 \leq a, b \leq X, \quad |c| \leq X,$$

that occur as Gram matrices of pairs in $\mathbb{Z}^3$. Then

$$N(X) \ll \frac{X^3}{\sqrt{\log X}}.$$

A proof is given in Appendix C. For orientation, if p is odd and $p \parallel D$, then a representable Gram matrix has rank 1 modulo p and its unique nonzero represented square class must be the class of -1 . Exactly half of the nonzero symmetric rank-1 matrices have the other square class. After lifting modulo p^2 , the resulting forbidden density is

$$\beta_p = \frac{(p^2 - 1)(p - 1)}{2p^4} = \frac{1}{2p} + O(p^{-2}). \quad (3)$$

A standard upper-bound sieve therefore has dimension $1/2$.

Theorem 2 (Sharp order of growth). *As $n \rightarrow \infty$,*

$$a(n) \asymp \frac{n^6}{\sqrt{\log n}}.$$

Consequently,

$$\log a(n) = 6 \log n - \frac{1}{2} \log \log n + O(1),$$

so that

$$\lim_{n \rightarrow \infty} \frac{\log a(n) - 6 \log n}{\log \log n} = -\frac{1}{2}, \quad \frac{a(n)}{n^6} \rightarrow 0.$$

Proof. For the lower bound, take the primitive pairs counted by $W(m)$. For each coordinate, the three integers $0, u_i, v_i$ have span at most $2m$, so the corresponding triangle can be translated coordinatewise into G_{2m} .

Fix one triangle congruence class. Choosing one of its three vertices as origin and ordering the other two gives at most $3! = 6$ ordered Gram matrices, all with the same determinant $D = 4A^2$, where A is the area. By Proposition 1, each such Gram matrix has exactly $24\rho(D)$ primitive ordered representations. Hence the total contribution of one congruence class to $W(m)$ is at most

$$6 \cdot 24 = 144.$$

Proposition 2 therefore gives

$$a(2m) \geq \frac{W(m)}{144} \gg \frac{m^6}{\sqrt{\log m}}.$$

Monotonicity from Section 3 extends this to all n .

For the upper bound, any triangle in G_n has edge vectors whose squared norms and mutual inner product have absolute value at most $3n^2$. Thus every congruence class supplies a represented positive definite Gram matrix in the box of Proposition 3 with $X = 3n^2$. Therefore

$$a(n) \leq N(3n^2) \ll \frac{n^6}{\sqrt{\log n}}.$$

Combining the two bounds proves the first assertion. The remaining statements follow by taking logarithms. $\square$

5 Perspective

The exact encoding by C_n and Theorem 2 reveal two different structures in the same problem. Coordinatewise, a triangle is built by adding three elements of the finite set C_n . Arithmetically, the resulting Gram matrices are filtered by local quadratic conditions, and those conditions account for the factor $1/\sqrt{\log n}$.

The theorem determines the sharp order of growth but not a leading constant. The available data are compatible with a limit of

$$\frac{a(n)\sqrt{\log n}}{n^6},$$

but proving such a limit would require finer information on how the different primitive embeddings of a fixed binary lattice sit inside an axis-aligned cube. We do not pursue that question here.

The same encoding also supports natural refinements. After sorting the squared side lengths $x \leq y \leq z$, right triangles are characterized by $x + y = z$, while $H(x, y, z) = 16A^2$ records the area. Thus angle classes and arithmetic conditions on the area lead to companion counting problems without changing the basic framework.

A Primitive Gram representations

We prove Proposition 1 and record the corresponding local existence criterion.

Let L be the rank-2 lattice with ordered basis e_1, e_2 and Gram matrix

$$G = \begin{pmatrix} a & c \\ c & b \end{pmatrix}, \quad D = ab - c^2 > 0,$$

and let $U = \mathbb{Z}^3$ with its usual dot product. A primitive ordered representation (u, v) is the same thing as a primitive isometric embedding

$$\iota : L \hookrightarrow U, \quad \iota(e_1) = u, \quad \iota(e_2) = v.$$

Put $w = u \times v$. The three coordinates of w , up to sign, are the 2×2 minors of the matrix with rows u, v . Smith normal form therefore shows that the embedding is primitive exactly when w is primitive. Moreover

$$\|w\|^2 = D,$$

so the orthogonal complement is the rank-1 lattice $K = \mathbb{Z}w$ of norm D .

For an integral lattice M , write

$$M^* = \{x \in M \otimes \mathbb{Q} : (x, M) \subseteq \mathbb{Z}\}, \quad A_M = M^*/M.$$

Pulling U back through the rational isometry

$$L \oplus K \longrightarrow U$$

gives an integral unimodular overlattice of index D . The usual gluing argument identifies it with the graph of an anti-isometry

$$\phi : A_L \longrightarrow A_K$$

of discriminant bilinear forms; see, for example, Conway and Sloane [3].

Conversely, every such anti-isometry produces a positive definite integral unimodular lattice U_ϕ of rank 3. This lattice is isometric to $\mathbb{Z}^3$. Indeed, Minkowski's convex-body theorem gives a nonzero vector of squared norm < 2 in any positive definite unimodular rank-3 lattice: a Euclidean ball of radius slightly larger than $(6/\pi)^{1/3}$ has volume greater than 8. Since squared norms are positive integers, there is a vector of norm 1. It splits off orthogonally, leaving a positive definite unimodular rank-2 lattice; the same argument finishes the decomposition.

Now

$$A_K \cong \mathbb{Z}/D\mathbb{Z},$$

and, for the generator $w/D + K$, its bilinear form is

$$(x, y) \longmapsto \frac{xy}{D} \pmod{\mathbb{Z}}.$$

If one anti-isometry exists, all anti-isometries form a torsor under the automorphism group of this discriminant form. Such an automorphism is multiplication by a unit $t \pmod{D}$, and preservation of the form is equivalent to

$$t^2 \equiv 1 \pmod{D}.$$

There are therefore exactly $\rho(D)$ anti-isometries.

For a fixed anti-isometry, choose one isometry $U_\phi \rightarrow U$. All others differ from it by an element of $O(\mathbb{Z}^3)$, the signed permutation group of order 48. Exactly half preserve the orientation determined by $w = u \times v$. Hence each anti-isometry gives exactly 24 ordered primitive representations. This proves Proposition 1.

For completeness we also record when an anti-isometry exists.

Lemma 2 (Local primitive representation criterion). *The matrix G has a primitive representation in $\mathbb{Z}^3$ if and only if*

1. $\gcd(a, b, c) = 1$;
2. $4 \nmid D$;
3. for every odd prime $p \mid D$, the unique nonzero square class represented by $G \pmod{p}$ is the class of -1 .

Equivalently, in (3), if $p \nmid a$ then

$$\left(\frac{-a}{p}\right) = 1,$$

and if $p \mid a$ then $p \nmid b$ and

$$\left(\frac{-b}{p}\right) = 1.$$

Proof. If a primitive embedding exists, then $A_L \cong A_K$ is cyclic of order D . The first Smith invariant of G is $\gcd(a, b, c)$, giving (1). Since $D = \|w\|^2$ for a primitive vector $w \in \mathbb{Z}^3$, at least one coordinate of w is odd, and a sum of three squares of a primitive vector is never 0 (mod 4); this gives (2).

Let $p^e \parallel D$ with p odd. Since G is primitive and singular modulo p , at least one diagonal entry is a p -adic unit. After a change of basis over $\mathbb{Z}_p$ we may write

$$L_p \cong \langle \varepsilon \rangle \perp \left\langle \frac{p^e d}{\varepsilon} \right\rangle, \quad D = p^e d, \quad p \nmid d,$$

with $\varepsilon \in \mathbb{Z}_p^\times$. The square class of $\varepsilon \bmod p$ is the unique nonzero square class represented by $G \bmod p$. The p -primary discriminant form of L has coefficient $\varepsilon d^{-1}/p^e$, while that of $K = \langle D \rangle$ has coefficient d/p^e . They are anti-isometric exactly when $-\varepsilon$ is a square in $\mathbb{Z}_p^\times$, which is condition (3).

Conversely, (1) makes A_L cyclic of order D . Condition (3) gives the required anti-isometry on every odd primary component. If $D \equiv 2 \pmod{4}$, the two 2-primary groups have order 2, and their unique nondegenerate bilinear forms are automatically anti-isometric because $-1/2 = 1/2$ modulo $\mathbb{Z}$. Thus the primary anti-isometries combine to $A_L \rightarrow A_K$, and the gluing construction above gives a primitive embedding into $\mathbb{Z}^3$. $\square$

B Weighted primitive pairs

We prove Proposition 2. Put

$$\Delta(u, v) = \|u \times v\|^2.$$

We first record the local densities used below.

Lemma 3. *For an odd prime p , the proportion of rank-2 pairs $(u, v) \in (\mathbb{F}_p^3)^2$ for which $p \mid \Delta(u, v)$ is*

$$\frac{p+1}{p^2+p+1}.$$

Every such zero is smooth, and for $e \geq 1$ the conditional density of $p^e \mid \Delta$ is

$$\frac{p+1}{p^2+p+1} p^{-(e-1)}.$$

For $p = 2$, under the rank-2 condition the corresponding densities are $3/7$ for $e = 1$ and 0 for $e \geq 2$.

Proof. There are

$$(p^3 - 1)(p^3 - p)$$

ordered rank-2 pairs. The condition $\Delta = 0$ says that their row plane is degenerate for $x_1^2 + x_2^2 + x_3^2$. Over $\mathbb{F}_p$ there are $p+1$ isotropic lines, and each determines one degenerate plane; each such plane has $|\mathrm{GL}_2(\mathbb{F}_p)|$ ordered bases. Hence the number of rank-2 zeros is

$$(p+1)|\mathrm{GL}_2(\mathbb{F}_p)| = p(p^2 - 1)^2.$$

Dividing gives the stated ratio. At a rank-2 zero, $u \times v$ is nonzero, so the gradient of $\Delta = (u \times v) \cdot (u \times v)$ does not vanish identically; equivalently the hypersurface is smooth there. Hensel lifting therefore contributes a factor $1/p$ for each additional power of p .

For $p = 2$, a primitive vector $w = u \times v$ has $\|w\|^2$ even exactly when two of its coordinates are odd, which occurs for 3 of the 7 nonzero vectors modulo 2. A primitive sum of three squares is never divisible by 4, so higher powers do not occur. $\square$

We also need a mild irreducibility fact.

Lemma 4. *The polynomial*

$$\Delta(u, v) = \|u\|^2 \|v\|^2 - (u \cdot v)^2$$

is absolutely irreducible.

Proof. Regard Δ as a quadratic polynomial in v_3 over

$$\mathbb{C}(u_1, u_2, u_3, v_1, v_2).$$

Its discriminant is

$$-4(u_1 v_2 - u_2 v_1)^2 (u_1^2 + u_2^2 + u_3^2).$$

The quadratic form $u_1^2 + u_2^2 + u_3^2$ has rank 3 and is not a square in the rational function field. Thus the discriminant is not a square, so the quadratic is irreducible. Gauss's lemma gives absolute irreducibility. $\square$

Set

$$Y = (\log m)^{3/4}, \quad Q_Y = \prod_{p \leq Y} p,$$

and let $\mathcal{A}_m$ be the ordered pairs in $[-m, m]^6$ whose 2×3 matrix has rank 2 modulo every prime $p \leq Y$, excluding the pairs with $\Delta = 0$. By the prime number theorem,

$$Q_Y = m^{o(1)}.$$

The local density of the rank-2 condition modulo p is

$$\delta_p = (1 - p^{-2})(1 - p^{-3}),$$

so the natural main term for $|\mathcal{A}_m|$ is

$$M(m) = (2m)^6 \prod_{p \leq Y} \delta_p \asymp m^6.$$

For $d \geq 1$, let $C_d(m)$ count the pairs in $\mathcal{A}_m$ for which $d \mid \Delta(u, v)$. By the Chinese remainder theorem there is a multiplicative density $h_m(d)$ such that, for $d \leq M(m)^\theta$ with a sufficiently small fixed $\theta > 0$,

$$C_d(m) = h_m(d) M(m) + O(M(m)^{1-\xi}) \quad (4)$$

for some fixed $\xi > 0$, uniformly in d and m . Indeed, all the congruence conditions live modulo

$$\text{lcm}(Q_Y, d) \leq m^{6\theta + o(1)},$$

and counting points in residue classes in a six-dimensional box gives an error $O(m^5 \text{lcm}(Q_Y, d) + m^{4+o(1)})$. The exact zero set $\Delta = 0$ has lower order and is absorbed in the same error.

For $p \leq Y$, Lemma 3 gives the local factors of h_m . For odd $p > Y$, a direct count gives

$$h_m(p) = \frac{p^5 + p^4 - p^3}{p^6} = \frac{1}{p} + \frac{1}{p^2} - \frac{1}{p^3}.$$

Thus, uniformly in m ,

$$h_m(p) = \frac{1}{p} + O(p^{-2}). \quad (5)$$

Moreover $h_m(p^e) \ll 1/p$. A standard prime-power zero estimate for an irreducible polynomial of degree 4 gives constants $C_0, C_1 > 0$ such that

$$h_m(p^e) \leq C_0 p^{-e/4+C_1};$$

see Pierce, Schindler and Wood [5, Lemma 4.10], as used in this context by Chan, Koymans, Pagano and Sofos [2, Corollary 1.14]. These are precisely the density hypotheses in [2, Definition 1.6], and (4) is stronger than the required level of distribution.

Apply their lower mean-value theorem [2, Theorem 1.13] to

$$f(r) = \frac{1}{\rho(r)}.$$

The function f is multiplicative, satisfies $0 < f \leq 1$, and is bounded away from zero on integers with bounded Ω . Since $\Delta(u, v) \leq 9m^4 \ll M(m)$, the growth condition in that theorem holds (for instance with exponent 1), and we obtain

$$\sum_{(u,v) \in \mathcal{A}_m} \frac{1}{\rho(\Delta(u, v))} \gg M(m) \prod_{p \leq M(m)} (1 - h_m(p)) \sum_{r \leq M(m)} \frac{h_m(r)}{\rho(r)}. \quad (6)$$

By (5) and Mertens' theorem,

$$\prod_{p \leq M} (1 - h_m(p)) \asymp \frac{1}{\log M}$$

uniformly in m . For the remaining sum it is enough to keep squarefree r . Put

$$g_m(p) = \frac{h_m(p)}{\rho(p)}.$$

For odd p ,

$$g_m(p) = \frac{1}{2p} + O(p^{-2}),$$

and the prime 2 changes only an absolute factor. Hence

$$Z_M := \prod_{p \leq M} (1 + g_m(p)) \asymp \sqrt{\log M}.$$

To see that a fixed proportion of this Euler-product mass already comes from $r \leq M$, choose the primes independently with probability $g_m(p)/(1 + g_m(p))$ and let R be their product. Then

$$\mathbb{E}(\log R) = \sum_{p \leq M} \frac{g_m(p)}{1 + g_m(p)} \log p = \left(\frac{1}{2} + o(1) \right) \log M.$$

Markov's inequality therefore gives $\Pr(R \leq M) \geq 1/3$ for all sufficiently large M . Consequently

$$\sum_{r \leq M} \frac{h_m(r)}{\rho(r)} \gg \sqrt{\log M}.$$

Substitution in (6) yields

$$\sum_{(u,v) \in \mathcal{A}_m} \frac{1}{\rho(\Delta(u, v))} \gg \frac{m^6}{\sqrt{\log m}}. \quad (7)$$

It remains to impose primitivity at primes larger than Y . A pair fails to be primitive at such a prime precisely when its 2×3 matrix has rank at most 1 modulo that prime. The

rank- ≤ 1 determinantal variety in $\mathbb{A}^6$ has codimension 2. The quantitative geometric sieve of Bhargava [1, Theorem 3.3] gives

$$\#\{(u, v) \in [-m, m]^6 : \text{rank}(u, v) \leq 1 \pmod{p} \text{ for some } p > Y\} \ll \frac{m^6}{Y \log Y} + m^5.$$

Because $Y = (\log m)^{3/4}$, this is

$$o\left(\frac{m^6}{\sqrt{\log m}}\right).$$

The weight $1/\rho$ is at most 1, so removing these pairs from (7) proves Proposition 2.

C The upper sieve

We prove Proposition 3. Let

$$G = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$$

be represented by a pair $u, v \in \mathbb{Z}^3$, and put $D = ab - c^2$. Fix an odd prime p with $p \nmid D$. Then $w = u \times v$ is nonzero modulo p and satisfies

$$w \cdot w = D \equiv 0 \pmod{p}.$$

Thus w is a nonzero isotropic vector. The Gram matrix $G \bmod p$ has rank 1, and its unique nonzero represented square class is the class of -1 ; this is also the local condition in Lemma 2.

There are $p^2 - 1$ nonzero symmetric rank-1 matrices over $\mathbb{F}_p$. Exactly half represent the other square class. Fix one such forbidden matrix modulo p . Among its p^3 lifts modulo p^2 , exactly $p^2(p - 1)$ have determinant divisible by p but not by p^2 : the coefficient of p in the determinant is a nonzero linear form in the three lifting parameters. Hence the forbidden density modulo p^2 is

$$\beta_p = \frac{(p^2 - 1)(p - 1)}{2p^4} = \frac{1}{2p} + O(p^{-2}),$$

as claimed in (3).

For squarefree d , combine these forbidden residue conditions modulo d^2 by the Chinese remainder theorem. Let $V \asymp X^3$ be the number of integer triples in the Gram box. If A_d denotes the number lying in the forbidden classes for every $p \mid d$, then elementary counting in residue classes gives

$$A_d = \beta(d)V + r_d, \quad \beta(d) = \prod_{p \mid d} \beta_p,$$

with

$$|r_d| \ll C^{\omega(d)}(X^2d + Xd^3 + d^5)$$

for an absolute constant C . Consequently, for $D_0 = X^{1/4}$,

$$\sum_{d \leq D_0} \mu^2(d) |r_d| = o\left(\frac{X^3}{\sqrt{\log X}}\right).$$

The standard upper-bound Selberg sieve, applied to the forbidden classes above, therefore gives

$$N(X) \ll X^3 \prod_{p < z} (1 - \beta_p)$$

for a fixed positive power z of X ; see, for example, Iwaniec and Kowalski [4, Chapter 6]. Since

$$\beta_p = \frac{1}{2p} + O(p^{-2}),$$

Mertens' theorem gives

$$\prod_{p < z} (1 - \beta_p) \asymp \frac{1}{\sqrt{\log z}} \asymp \frac{1}{\sqrt{\log X}}.$$

This proves Proposition 3.

Declaration on AI-assisted tools

AI-assisted tools were used during mathematical exploration, drafting, and language editing. The author checked the mathematical statements and proofs and takes full responsibility for the content.

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